A smooth counterexample to critical–subcritical heat-kernel comparison

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Abstract

We construct two nonnegative Schrödinger operators on the line with smooth bounded potentials and spectral bottom zero. One is subcritical and the other null-critical, while the quotient of their diagonal heat kernels has infinite limsup. This gives a negative answer to the heat-kernel comparison question without an ordering assumption on the potentials. The construction preserves small heat-kernel values of successive operators with positive spectral gaps, then restores criticality by adding distant intervals on which a positive zero-energy solution equals one.

1 The comparison question

The distinction between critical and subcritical Schrödinger operators can be read from the time integral of the positive heat kernel. It is natural to ask whether this distinction also orders their decay at large times. Fraas, Krejčiřík and Pinchover formulated such a comparison in [1, Conjecture 1]; its self-adjoint heat-kernel version appears again in [2, Conjecture 2]. Their conditional results include an additional global heat-kernel comparison hypothesis [1, Theorem 3.1] and nonnegative potential perturbations [1, Corollary 1].

We consider the formulation in AIM Problem 217 [3]. On a connected, complete, noncompact smooth Riemannian manifold without boundary, let $H_j = -\Delta_g + V_j$, $j \in \{0, +\}$, be the Friedrichs operators for smooth real potentials bounded below. Assume $H_j \geq 0$ and $\inf \sigma(H_j) = 0$. Write $k_j(t, x, y) > 0$ for the heat kernel of e^{-tH_j}. At some fixed distinct points x_*, y_*, suppose that

$$\int_0^\infty k_+(t, x_*, y_*)\, \mathrm{d}t < \infty, \qquad \int_0^\infty k_0(t, x_*, y_*)\, \mathrm{d}t = \infty, \qquad \ker_{L^2} H_0 = \{0\}. \tag{1}$$

These are the subcritical and null-critical cases, respectively. With no pointwise ordering of V_+ and V_0, the question is whether

$$\lim_{t\to\infty} \sup_{(x,y)\in K} \frac{k_+(t, x, y)}{k_0(t, x, y)} = 0 \quad \text{for every compact } K \subset M \times M. \tag{2}$$

Theorem 1.1. *There are real potentials $V_+, V_0 \in C^\infty(\mathbb{R}) \cap L^\infty(\mathbb{R})$ such that their Friedrichs operators $H_j = -\mathrm{d}^2/\mathrm{d}x^2 + V_j$ on $L^2(\mathbb{R}, \mathrm{d}x)$ satisfy*

$$H_j \geq 0, \qquad \inf \sigma(H_j) = 0, \qquad \ker_{L^2} H_0 = \{0\},$$

and their positive heat kernels satisfy

$$\int_0^\infty k_+(t, 0, 1)\, \mathrm{d}t < \infty, \qquad \int_0^\infty k_0(t, 0, 1)\, \mathrm{d}t = \infty, \tag{3}$$

but

$$\limsup_{t\to\infty} \frac{k_+(t, 0, 0)}{k_0(t, 0, 0)} = \infty. \tag{4}$$

The Euclidean line is an admissible manifold. The compact singleton $K = \{(0,0)\}$ therefore disproves (2); the distinctness of the points in (1) imposes no restriction on this K.

The numerator will be a fixed nonzero nonnegative compactly supported bump. Its diagonal heat kernel is bounded below by $ct^{-3/2}$. For the reference operator we use a positive zero-energy solution $h = e^w$ and set $V_0 = w'' + (w')^2$. A finite construction with $w' = 1$ on its positive tail has a strictly positive spectral gap, so its heat kernel is eventually much smaller than $ct^{-3/2}$. We choose one such time, then a radius so large that later changes outside that radius preserve the estimate. Beyond the radius we let w return to zero on a unit interval. Repeating this produces one fixed potential with infinitely many such intervals. They make the reference Green integral infinite without losing the selected small kernel values. The analytic estimates are given in Section 2; Section 3 carries out the construction. The one-dimensional Dirichlet and Green-kernel justification is proved in Appendix A.

2 Heat-kernel estimates

For a bounded real continuous potential U, write $H_U = -\mathrm{d}^2/\mathrm{d}x^2 + U$. Its self-adjoint domain is $H^2(\mathbb{R})$, its form domain is $H^1(\mathbb{R})$, and its Friedrichs form is

$$Q_U[u] = \int_{\mathbb{R}} \left(|u'|^2 + U|u|^2\right) \mathrm{d}x. \tag{5}$$

These identifications follow from the free operator and bounded perturbation [5, Theorems 6.4 and 7.8, Lemma 7.9]. One can check the form core directly: first multiply an H^1 function by smooth cutoffs whose derivatives are $O(R^{-1})$, then mollify. Both the function and its weak derivative converge in L^2, so $C_c^\infty(\mathbb{R})$ is a form core.

Our free kernel and Brownian-bridge convention are

$$\begin{aligned} p(t,x,y) &= (4\pi t)^{-1/2} e^{-(x-y)^2/(4t)}, \\ k_U(t,x,y) &= p(t,x,y)\, \mathbb{E} \exp\left(-t \int_0^1 U(b_r)\, \mathrm{d}r\right), \\ b_r &= (1-r)x + ry + \sqrt{2t}\, \beta_r, \end{aligned} \tag{6}$$

where β is the standard unit-time Brownian bridge. This is the Feynman–Kac formula [4, Theorem 1.1], applied to $U/2$ at time $2t$. In particular, k_U is strictly positive and continuous for $t > 0$. If $U \geq -B$, then $k_U \leq e^{Bt} p$. The symmetry and semigroup identities hold for these continuous representatives, and their convolution integrals are controlled by the Gaussian bound.

Lemma 2.1. *If bounded continuous real potentials $U, W \geq -B$ agree on $[-R, R]$, then*

$$|k_U(t,0,0) - k_W(t,0,0)| \leq 4e^{Bt}(4\pi t)^{-1/2} e^{-R^2/t}, \qquad t > 0. \tag{7}$$

Proof. Reflection of Brownian paths at their first hit of a level preserves Wiener law [6, Proposition 4]. For the variance-two bridge from x to y, with $x, y < a$, division of the reflected endpoint density by the free endpoint density gives

$$\mathbb{P}\{\max b_r \geq a\} = \frac{p(t,x,2a-y)}{p(t,x,y)} = e^{-(a-x)(a-y)/t}. \tag{8}$$

The lower-level formula follows by negating the path. A bridge from zero to zero thus exits $[-R, R]$ with probability at most $2e^{-R^2/t}$. On paths that do not exit, the two weights in (6) coincide; on the remaining paths their difference has absolute value at most $2e^{Bt}$. Multiplying these bounds by $p(t,0,0)$ proves (7). □

We also need an exact formula for the time-integrated kernel. We state it here and prove the Dirichlet identification and exhaustion in Appendix A, so that the construction can be read separately from that operator argument.

Lemma 2.2. *Let U be smooth and bounded on $\mathbb{R}$, and let $h > 0$ solve $-h'' + Uh = 0$. Put $s(x) = \int_0^x h(r)^{-2}\,\mathrm{d}r$. For $a > \max(|x|, |y|)$, define*

$$G_a(x,y) = h(x)h(y)\frac{[s(\min(x,y)) - s(-a)][s(a) - s(\max(x,y))]}{s(a) - s(-a)}. \tag{9}$$

Then

$$\int_0^\infty k_U(t,x,y)\,\mathrm{d}t = \lim_{a\to\infty} G_a(x,y), \tag{10}$$

where the limit may be infinite and may be taken along increasing integers. If $D = (a,\infty)$ and $U = 0$ on D, its Dirichlet kernel is

$$p_D(t,u,v) = p(t,u,v) - p(t,u,2a-v), \qquad u,v > a, \tag{11}$$

and $0 \le p_D(t,u,v) \le k_U(t,u,v)$.

Lemma 2.3. *Set*

$$q(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1, \\ 0, & |x| \ge 1. \end{cases}$$

For $H_+ = H_q$, one has $H_+ \ge 0$, $\inf\sigma(H_+) = 0$ and $\int_0^\infty k_+(t,0,1)\,\mathrm{d}t < \infty$. There is a fixed $c > 0$ such that

$$k_+(t,0,0) \ge ct^{-3/2} \qquad (t \ge 3). \tag{12}$$

Proof. The bump is smooth, even and nonnegative, so $H_+ \ge 0$. Take $\psi \in C_c^\infty((1,2))$ with $\|\psi\|_2 = 1$ and set $\psi_L(x) = L^{-1/2}\psi((x-2)/L)$ for $L \ge 1$. These functions are supported where $q = 0$ and have Rayleigh quotient $L^{-2}\|\psi'\|_2^2$. Hence $\inf\sigma(H_+) = 0$.

Let $h_+'' = qh_+$, with $h_+(0) = 1$, $h_+'(0) = 0$. For completeness, the Volterra equation

$$h_+(x) = 1 + \int_0^x (x-r)q(r)h_+(r)\,\mathrm{d}r \quad (x \ge 0)$$

constructs this solution: its mth iterated term is bounded by $\|q\|_\infty^m x^{2m}/(2m)!$ on compact intervals. The series and its differentiated series converge there, and the same bound for a difference gives uniqueness. All terms are nonnegative. Thus $h_+ \ge 1$, $h_+' \ge 0$, and $b = h_+'(1) > 0$. Even reflection gives the global smooth positive solution. If $A = h_+(1)$, then for $x \ge 1$,

$$h_+(x) = A + b(x-1), \qquad S_+ := \int_0^\infty h_+^{-2} = \int_0^1 h_+^{-2} + \frac{1}{bA} < \infty.$$

In (9), s is odd and tends to $\pm S_+$ at the two ends. Every factor at $(x,y) = (0,1)$ has a finite limit, and the denominator tends to $2S_+ > 0$. Equation (10) proves the finite integral.

To obtain a lower bound, let $I = [2,3]$, $D = (1,\infty)$ and $m = \min_{u\in I} k_+(1,0,u) > 0$. The semigroup identity, positivity, and Lemma 2.2 give, for $s = t - 2 \ge 1$,

$$k_+(t,0,0) \ge m^2 \int_I\int_I p_D(s,u,v)\,\mathrm{d}u\,\mathrm{d}v.$$

On I^2, the image formula becomes

$$p_D(s,u,v) = (4\pi s)^{-1/2}e^{-(u-v)^2/(4s)}\big[1 - e^{-(u-1)(v-1)/s}\big].$$

Here $|u-v| \le 1$, $(u-1)(v-1) \ge 1$, and concavity gives $1 - e^{-1/s} \ge (1-e^{-1})/s$. Since $s \le t$ and I^2 has area one, (12) follows with

$$c = \frac{m^2 e^{-1/4}(1-e^{-1})}{\sqrt{4\pi}} > 0. \tag{13}$$

□

3 Construction of the reference potential

We use one fixed smooth step

$$\theta(r) = \frac{e^{-1/r}}{e^{-1/r} + e^{-1/(1-r)}} \quad (0 < r < 1), \qquad \theta = 0 \ (r \le 0), \quad \theta = 1 \ (r \ge 1). \tag{14}$$

It is flat at both endpoints, strictly increasing on $(0,1)$, and satisfies $\theta(1-r) = 1 - \theta(r)$. In particular, $\int_0^1 \theta = 1/2$ and $B := \|\theta'\|_\infty < \infty$. Consider smooth, even, nonnegative functions w with

$$|w'| \le 1, \qquad |w''| \le B. \tag{15}$$

For each such function set $U_w = w'' + (w')^2$ and $h = e^w$. Direct integration by parts on compactly supported smooth functions gives

$$Q_{U_w}[u] = \|u' - w'u\|_2^2 \ge 0. \tag{16}$$

Indeed, the cross term is $-\int w'(|u|^2)' = \int w''|u|^2$, also for complex u. The bounds (15) and the form-core approximation extend (16) to $H^1(\mathbb{R})$. Thus $H_{U_w} \ge 0$ and $-h'' + U_w h = 0$. Any L^2 zero mode belongs to H^2 and satisfies $u' - w'u = 0$, whence $(u/h)' = 0$ on each compact interval and $u = Ch$. In particular, $h \notin L^2$ excludes every nonzero L^2 zero mode.

Lemma 3.1. *If, in addition, $w' = 1$ for all sufficiently large positive x, then $\lambda(w) := \inf \sigma(H_{U_w}) > 0$. For $t \ge 1$,*

$$k_{U_w}(t,0,0) \le M e^{-\lambda(w)(t-1)}, \qquad M = \frac{e^B}{\sqrt{4\pi}}. \tag{17}$$

Proof. Evenness implies $U_w = 1$ outside a compact interval, and e^w grows exponentially at both ends. If the spectral bottom were zero, the spectral form characterization [5, Theorem 3.7 and equations (3.50)–(3.51)] would give $u_\ell \in H^1$ with $\|u_\ell\|_2 = 1$ and $Q_{U_w}[u_\ell] \to 0$. Since $U_w \ge -B$, this sequence is bounded in H^1. Pass to a weakly convergent subsequence there [5, Lemma 1.13], with limit u, and to a further subsequence converging strongly in L^2 on every compact interval. The last assertion can be checked in one dimension as follows. On any fixed interval,

$$|v(x) - v(y)| \le \|v'\|_2 |x-y|^{1/2};$$

together with a point where $|v|$ is at most its L^2 average, this gives uniform bounds and a common modulus of continuity. Successively finer finite grids give a subsequence converging uniformly on that interval. A diagonal choice over $[-j, j]$, $j \in \mathbb{N}$, proves the stated local convergence.

If $u = 0$, compact support and boundedness of $U_w - 1$ would give

$$Q_{U_w}[u_\ell] = \|u_\ell'\|_2^2 + 1 + \int (U_w - 1)|u_\ell|^2 \ge 1 + o(1),$$

a contradiction. Thus $u \ne 0$. But (16) implies $u_\ell' - w'u_\ell \to 0$ strongly in L^2, so its weak limit is $u' - w'u = 0$. This forces $u = Ce^w$, impossible for a nonzero L^2 function. The gap is therefore strictly positive.

For $f(z) = k_{U_w}(1/2, z, 0)$, the Gaussian bound gives $f \in L^2$ and $\|f\|_2^2 = k_{U_w}(1,0,0)$. The semigroup identity and spectral bound give

$$k_{U_w}(t,0,0) \le e^{-\lambda(w)(t-1)} \|f\|_2^2 \le e^{-\lambda(w)(t-1)} e^B (4\pi)^{-1/2}.$$

The constant M is common to these functions; their positive gaps need not have a common lower bound. □

The next extension lets a function with an outward slope return to height zero. Suppose R lies on the slope-one tail and $a = w(R) \geq 1/2$. Keep w on $[-R, R]$ and prescribe its derivative on the positive side as follows:

1. On $[R, R+2]$, use $1 - 2\theta((x-R)/2)$. The slope turns from one to minus one, with total height change zero; throughout this turn the height is at least a.
2. Continue with slope -1 for length $a - 1/2$, reaching height $1/2$.
3. On the next unit interval, with local coordinate $r \in [0,1]$, use derivative $-1 + \theta(r)$. The height decreases by $1/2$ and reaches zero.
4. Keep height zero on a unit interval J.
5. On the next unit interval use derivative $\theta(r)$, reaching height $1/2$ with slope one, and continue with slope one thereafter.

Every piece stays nonnegative. Flatness of θ matches all jets, even when the straight descent has zero length. The second derivative in the first turn is $-\theta'((x-R)/2)$ and in each unit turn has absolute value at most B. Thus (15) is preserved. Reflect on the negative side; the central part is unchanged. Denote this extension by w_{new}.

Proof of Theorem 1.1. Fix $V_+ = q$ and the constant c of Lemma 2.3. Start from

$$w_1(x) = \int_0^{|x|} \theta(r-1)\,\mathrm{d}r.$$

This function is smooth and zero on $[-1,1]$, obeys (15), and has positive-side slope one for $x \geq 2$. Set $t_0 = R_0 = 0$.

Given w_n, let $U_n = w_n'' + (w_n')^2$ and $\lambda_n = \inf \sigma(H_{U_n}) > 0$, using Lemma 3.1. First choose an integer t_n such that

$$t_n \geq \max(n+3, t_{n-1}+1), \qquad M e^{-\lambda_n(t_n-1)} \leq \frac{c}{2n} t_n^{-3/2}. \tag{18}$$

This is possible because λ_n is positive for this fixed w_n. Next choose an integer R_n on its slope-one tail, beyond every earlier valley, with $R_n > \max(n, R_{n-1}+1)$ and $w_n(R_n) \geq 1$, such that

$$4e^{Bt_n}(4\pi t_n)^{-1/2} e^{-R_n^2/t_n} \leq \frac{c}{2n} t_n^{-3/2}. \tag{19}$$

Take R_n at least one unit beyond the beginning of that tail as well. All these conditions involve finitely many fixed quantities, and the left side of (19) tends to zero as R_n increases. Apply the extension just described at R_n to obtain w_{n+1} and its new unit zero interval J_n. If desired, taking the first admissible integers at both choices makes the construction fully specified without a numerical evaluation of λ_n.

We have $w_{n+1} = w_n$ on $[-R_n, R_n]$, with $R_n \to \infty$. Every compact neighborhood is therefore eventually unchanged, with all derivatives. Its eventual value defines one smooth even nonnegative function w obeying (15). It is zero on $[-1,1]$ and on all the disjoint unit intervals J_n and $-J_n$. Define the single final potential and solution

$$V_0 = w'' + (w')^2, \qquad h_0 = e^w. \tag{20}$$

Then $-B \leq V_0 \leq B+1$, $H_0 \geq 0$ by (16), and $\ker_{L^2} H_0 = \{0\}$ because $h_0 \geq 1$.

The preserved unit zero intervals give

$$\int_0^\infty h_0^{-2} = \int_{-\infty}^0 h_0^{-2} = \infty. \tag{21}$$

Put $s(x) = \int_0^x h_0^{-2}$ and $S = s(a)$. Evenness makes s odd, and $S \to \infty$ as $a \to \infty$. On the central interval $h_0 = 1$, so $s(1) = 1$. For $a > 1$, (9) at $(0, 1)$ reads

$$G_a(0, 1) = \frac{S - 1}{2}.$$

Lemma 2.2 proves the infinite reference integral in (3). The finite numerator integral has already been proved at the same distinct pair.

For the spectral bottom, use the compactly supported H^1 function

$$u_a(x) = h_0(x) \max\left(0, 1 - \frac{|s(x)|}{S}\right).$$

It vanishes outside $[-a, a]$, and its zero endpoint values ensure that its zero extension is in H^1. Equation (16) yields

$$Q_0[u_a] = \frac{1}{S^2} \int_{-a}^{a} h_0^{-2} \, \mathrm{d}x = \frac{2}{S}.$$

If $S \geq 2$, its square is at least $1/4$ on $[-1, 1]$, whence $\|u_a\|_2^2 \geq 1/2$. The Rayleigh quotients are at most $4/S \to 0$; together with nonnegativity, this gives $\inf \sigma(H_0) = 0$.

Finally, $V_0 = U_n$ on $[-R_n, R_n]$, and both have the same lower bound $-B$. Lemmas 2.1 and 3.1, followed by (18)–(19), give

$$0 < k_0(t_n, 0, 0) \leq k_{U_n}(t_n, 0, 0) + \frac{c}{2n} t_n^{-3/2} \leq \frac{c}{n} t_n^{-3/2}.$$

The same fixed c appears in (12). Consequently

$$\frac{k_+(t_n, 0, 0)}{k_0(t_n, 0, 0)} \geq n,$$

which proves (4). Both potentials are fixed; only the comparison times tend to infinity. □

The constructed potentials are unordered. At the middle of each length-two downward turn, $w' = 0$ and $w'' = -\theta'(1/2) < 0$, so $V_0 < 0$ outside the support of q. On a retained slope-one segment $V_0 = 1$ while $q = 0$. Thus $q - V_0$ takes both signs, as is consistent with the nonnegative perturbation result in [1, Corollary 1].

A Dirichlet kernels and Green functions

We prove Lemma 2.2, including the identification of killed paths with the one-dimensional Friedrichs Dirichlet operator. Only the whole-line formula (6) is imported from [4]. The form, weak compactness, spectral calculus and uniform polynomial approximation used below are given in [5, Theorems 2.12–2.13, Lemma 1.13, Theorems 3.1, 3.7 and 1.15].

The killed semigroup

Let I be a finite open interval or a half-line, and let $U \geq -B$ be bounded, real and continuous on $\mathbb{R}$. Define H_U^I on $L^2(I)$ by the closed form $Q_I[u] = \int_I (|u'|^2 + U|u|^2)$ with domain $H_0^1(I)$. Set

$$F(x) = \min\{1, \operatorname{dist}(x, \overline{I})^2\}, \quad H_m = H_{U+mF}, \quad \alpha = B + 1.$$

The shifted forms

$$a_m[u] = \int_{\mathbb{R}} (|u'|^2 + (U + \alpha)|u|^2 + mF|u|^2) \geq \|u\|_{H^1}^2$$

are coercive. For $f \in L^2(\mathbb{R})$, put $u_m = (H_m + \alpha)^{-1} f$. Testing its variational equation with u_m gives

$$\|u_m\|_{H^1} \le \|f\|_2, \qquad m \int F|u_m|^2 \le \|f\|_2^2.$$

A weak H^1 subsequential limit u vanishes outside $\overline{I}$, since on every compact set where $F \ge \delta > 0$ the local L^2 norm tends to zero. An H^1 function in one dimension is locally absolutely continuous: integrate its weak derivative and subtract, leaving a distribution with derivative zero. Thus u has zero trace at every finite endpoint of I.

These zero extensions are precisely $H_0^1(I)$. To check the needed density at an endpoint a, use $u(a) = 0$ and

$$|u(x)|^2 \le |x - a| \int_{\min(a,x)}^{\max(a,x)} |u'|^2.$$

A cutoff vanishing within distance δ of a and equal to one beyond 2δ has derivative-of-cutoff error bounded by a constant times $\int_{|x-a|<2\delta} |u'|^2$, which tends to zero. Ordinary cutoffs at infinity, followed by mollification away from the endpoints, complete the H^1 approximation. Translation continuity in L^2 justifies convergence for both u and u'; equivalently this follows from the Fourier and weak derivative description [5, Section 7.1, (7.12)–(7.15)].

Tests compactly supported in I remove the term mF from the variational equation. Passing to the weak limit and using this density identifies $u = (H_U^I + \alpha)^{-1}(f|_I)$, extended by zero. The solution is unique, so every weak subsequential limit is the same. Since $Fu = 0$, the variational equations give the exact identity

$$a_m[u_m - u] = \mathrm{Re}\langle f, u_m - u\rangle \longrightarrow 0.$$

Hence $u_m \to u$ strongly in H^1 and L^2. In particular, the bounded positive resolvents $R_m = (H_m + \alpha)^{-1}$ converge strongly to the Dirichlet resolvent on $L^2(I)$, extended by zero on $L^2(I^c)$; call this limit R_∞.

The limiting resolvent vanishes on $L^2(I^c)$, so we pass to semigroups through bounded continuous calculus. For fixed $t > 0$, the function

$$f_t(r) = e^{-t(1/r - \alpha)} \quad (r > 0), \qquad f_t(0) = 0$$

is continuous on $[0, 1]$. The spectra of R_m and R_∞ lie in that interval. Strong convergence and uniform boundedness give convergence of every finite power. Uniform polynomial approximation then gives $f_t(R_m) \to f_t(R_\infty)$ strongly. The spectral calculus identifies this as convergence of e^{-tH_m} to $e^{-tH_U^I}$ extended by zero.

For the passage from the penalized kernels, recall the variance-two bridge and its free prefactor:

$$b_r = (1 - r)x + ry + \sqrt{2t}\,\beta_r, \qquad p(t, x, y) = (4\pi t)^{-1/2} e^{-(x-y)^2/(4t)},$$

where β is the standard unit-time bridge. In the full-line formula (6) for $U + mF$, the weights decrease to

$$\exp\left(-t \int_0^1 U(b_r)\,\mathrm{d}r\right) \mathbf{1}_{\{b_r \in \overline{I} \text{ for all } r\}}.$$

Continuity of each path identifies the zero integral of F with this event. For an upper level $z > \max(x, y)$, the reflection formula (8) reads

$$\mathbb{P}\{\max_{0 \le r \le 1} b_r \ge z\} = e^{-(z-x)(z-y)/t}.$$

Its continuity in z excludes atoms of the maximum above both endpoints; negating the path gives the corresponding statement for the minimum. Thus replacing $\overline{I}$ by I changes a null

event for interior endpoints. The Gaussian bound allows passage to the limit against compactly supported L^2 tests, identifying the strong semigroup limit. We obtain the killed kernel

$$k_U^I(t,x,y) = p(t,x,y)\,\mathbb{E}\left[e^{-t\int_0^1 U(b_r)\,\mathrm{d}r}\mathbf{1}_{\{b_r\in I \text{ for all } r\}}\right]. \tag{22}$$

This identity is pointwise: coupling bridges with the same β proves continuity at interior (t,x,y) by dominated convergence, since failure of convergence of the survival indicator requires a boundary maximum or minimum. In particular, $0 \le k_U^I \le k_U$ and $k_U^I \le e^{Bt}p$. For $I = (a,\infty)$ and $U = 0$ on I, reflection gives the image formula (11). For $I = (-a,a)$ increasing to $\mathbb{R}$, every bridge has bounded range, and its survival indicators increase to one. Hence

$$k_U^{(-a,a)}(t,x,y) \uparrow k_U(t,x,y) \quad \text{at each fixed } t>0, x, y. \tag{23}$$

The interval inverse and its time integral

Now suppose U is smooth and bounded, and $h > 0$ solves $-h'' + Uh = 0$. On $(-a,a)$ put $m_a = \min h > 0$, $M_a = \max h$, and $s(x) = \int_0^x h^{-2}$. Multiplication and division by h preserve $H_0^1(-a,a)$. For $u = hv$, integration by parts and density give

$$Q_I[u] = \int_I h^2|v'|^2 \ge \frac{m_a^2}{4a^2M_a^2}\|u\|_2^2.$$

Here $\|v\|_2^2 \le 4a^2\|v'\|_2^2$ follows by integrating $v(x) = \int_{-a}^x v'$. Thus the Dirichlet operator has a positive gap $\lambda_a = m_a^2/(4a^2M_a^2)$, allowed to depend on a.

The homogeneous solutions $\ell(x) = h(x)[s(x) - s(-a)]$ and $r(x) = h(x)[s(a) - s(x)]$ satisfy

$$\ell' r - \ell r' = s(a) - s(-a) > 0.$$

Their Green function is

$$G_a(x,y) = \frac{\ell(\min(x,y))r(\max(x,y))}{s(a) - s(-a)},$$

which is exactly (9). It is continuous, zero at the endpoints, and its x derivative has jump -1 at $x = y$. Thus $H_U^I G_a(\cdot,y) = \delta_y$. Integration against a smooth compactly supported function gives the Dirichlet inverse, whose uniqueness follows from the positive gap.

Spectral integration on $[\lambda_a,\infty)$ gives $\int_0^\infty e^{-tH_U^I}\,\mathrm{d}t = (H_U^I)^{-1}$. The kernel equality also holds at every point. For $0 < t \le 1$, (22) is bounded by $e^B(4\pi t)^{-1/2}$, integrable at zero in dimension one. For $t \ge 1$, the time-1/2 kernel vectors and the spectral bound give

$$|k_U^I(t,x,y)| \le e^{-\lambda_a(t-1)}\sqrt{k_U^I(1,x,x)k_U^I(1,y,y)} \le \frac{e^B}{\sqrt{4\pi}}e^{-\lambda_a(t-1)}.$$

These bounds make the time-integrated kernel continuous, including on the diagonal. Its equality with the inverse kernel, initially almost everywhere, therefore holds everywhere. Finally, monotone convergence in (23) gives

$$\int_0^\infty k_U(t,x,y)\,\mathrm{d}t = \lim_{a\to\infty}\int_0^\infty k_U^{(-a,a)}(t,x,y)\,\mathrm{d}t = \lim_{a\to\infty} G_a(x,y),$$

with infinity allowed. This proves Lemma 2.2.

AI-use statement

OpenAI's Codex assisted in developing and writing the argument, checking cited passages, and conducting internal AI reviews of the proof and manuscript. The mathematical claims are submitted for independent review.

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